Supporting Information

It depends how you count: definition, disturbance history, and the distribution of late-successional and old-growth forest across the northeastern United States

Aaron R. Weiskittel, University of Maine, Center for Research on Sustainable Forests and School of Forest Resources

Table S1. v5.1 six-dimension structural proxy: dimension definitions and scoring.

Dim	Variable	Source	1 point	2 points
1	Large-tree basal area	FIA trees, DBH ≥ 20 in	≥ 40 ft²/ac	≥ 80 ft²/ac
2	Stand maturity	STDAGE (max DBH fallback)	≥ 80 yr (or DBH ≥ 24 in)	≥ 120 yr
3	Structural diversity	TPA-weighted SD of DBH	≥ 5 in	≥ 8 in
4	Total stocking	Total live basal area	≥ 100 ft²/ac	≥ 150 ft²/ac
5	Deadwood	Standing snag TPA (DBH ≥ 5 in)	≥ 75th pct	≥ 90th pct
6	Canopy height	Potapov 2021 RH95, 30 m	≥ 18 m	≥ 25 m

Table S2. Integrated v5.1 LSOG share by state, 2019–2023 panel (design-based, 95% CI).

State	n plots	Any-LSOG % [95% CI]	OG-class %
Maine	3,125	14.1 [12.9, 15.3]	0.19
New York	2,107	27.2 [25.2, 29.0]	1.5
Vermont	657	28.8 [25.4, 32.3]	0.30
New Hampshire	757	31.2 [28.0, 34.3]	0.66

Table S3. Design-based older forest by axis across the Northeast (2024, % of forestland), showing axes rank states differently.

State	Stand age ≥ 120 yr % [95% CI]	Large-tree BA ≥ 30 ft²/ac % [95% CI]
Maine	3.9 [3.3, 4.6]	12.5 [11.4, 13.7]
New Hampshire	1.8 [0.9, 2.6]	37.9 [34.7, 41.0]
Vermont	1.0 [0.4, 1.6]	39.1 [35.8, 42.4]
Massachusetts	1.5 [0.4, 2.5]	56.1 [51.6, 60.7]
Connecticut	6.8 [3.8, 9.8]	57.7 [51.6, 63.8]
Rhode Island	1.2 [0.0, 3.1]	47.0 [37.9, 56.1]

Table S4. Dimensionality of LSOG attributes (Maine plots): selected pairwise correlations and PCA variance.

Attribute pair / component	Value
Large-tree BA vs large-tree count	r = 0.98
Large-tree BA vs diameter diversity	r = 0.77
Large-tree BA vs standing dead BA	r = 0.51
Large-tree BA vs coarse woody debris	r = 0.48
Standing dead BA vs coarse woody debris	r = 0.47
PC1 variance explained	58.8%
PC1-PC2 cumulative	72.6%
PC1-PC4 cumulative	90.8%

Table S5. LSOG occurrence rate by harvest-probability and slope tercile.

Tercile	LSOG rate by harvest probability %	LSOG rate by slope %
Low	13.1	11.8
Medium	24.1	21.7
High	33.5	37.3

Table S6. Representation of true LSOG (all four axes, LCMS continuity) across forest-type groups, pooled across the four states (design-based).

Forest-type group	Total forest (K ac)	True LSOG % [95% CI]	True LSOG (K ac)
Maple/beech/birch (northern hardwood)	22,995	11.6 [10.6, 12.7]	2,676
Spruce/fir	7,428	2.1 [1.3, 2.8]	153
Oak/hickory	4,432	6.6 [4.7, 8.6]	294
White/red/jack pine	3,387	21.2 [17.2, 25.3]	720
Aspen/birch	2,527	0.6 [0.0, 1.3]	15
Elm/ash/cottonwood	1,772	0.0 [0.0, 0.0]	0
Oak/pine	1,475	16.4 [10.8, 22.0]	242
Other softwood	258	1.8 [0.0, 5.0]	5

Table S7. Global (variance-based Sobol) sensitivity of the multi-model ensemble LSOG probability to the two predictors, over their empirical ranges. S1 is the first-order index (main effect); ST is the total-order index (main plus interactions). Canopy height dominates; the disturbance axis contributes mainly through interaction. (S1 for canopy height slightly exceeds 1 owing to Monte Carlo estimator noise and is interpreted as approximately unity.)

Predictor	First-order S1	Total-order ST
Canopy height (Potapov/GEDI)	~1.0	0.97
Time since disturbance (LCMS)	0.08	0.12

Table S8. Mean mapped LSOG probability by learner in the multi-model ensemble (Maine, forested cells). The wide spread is the model-structural uncertainty mapped in Fig. 7b.

Learner	Mean mapped P(LSOG)
Balanced random forest	0.50
Probability forest (ranger)	0.26
Weighted logistic	0.59
Support vector machine	0.27
Multi-objective SVR	0.26
Ensemble mean	0.37

Table S9. Rare-class remedies for the reproduced Hagan random forest, with old-growth (OG) as the rare event. Three standard corrections and a voting-threshold adjustment are compared against the default. OG operating accuracy (recall) and the wall-to-wall mapped OG area both move substantially, confirming that the headline area is a modeling choice rather than a fixed quantity. (The OG class rests on only 17 training hectares, so OG-specific accuracies are themselves highly uncertain; plot-level bootstrap intervals are too wide to be informative.)

Strategy (scikit-learn / imblearn analog)	OG recall	Overall acc.	Mapped OG area (%)
Default, unbalanced	0.24	0.87	1.0
Class weighting (classwt; class_weight='balanced')	0.29	0.86	0.8
Balanced subsample (sampsize/strata; BalancedRandomForestClassifier)	0.71	0.84	1.9
Voting-threshold adjustment (OG vote ≥ 0.2)	0.82	0.87	2.3

Table S10. Final robustness summary: each headline conclusion and the analytical perturbation it survives.

Conclusion	Stress applied	Result	Robust
Maine carries the lowest LSOG share	scoring threshold, drop canopy dimension, FIA panel	lowest in 8 of 9 tests	yes
True LSOG is rare (3 to 5% in Maine)	five LCMS continuity definitions	3.1 to 5.2%	yes
Credible maps disagree	three independent remote-sensing products	1.6 to 2.6 fold, agree on 2.7%	yes

Half the plot-level cross-map disagreement is geolocation	fuzzed vs true FIA coordinates	kappa 0.13 to 0.29	yes
Rare-class detection and area are choice-driven	default vs weighted vs balanced vs threshold RF	OG recall 0.24 to 0.82; area 1.0 to 2.3%	yes
Model structure dominates map uncertainty	five-learner ensemble	mean P 0.26 to 0.59, SD 0.18	quantified
Canopy height drives the signal	variance-based Sobol indices	first-order ~1.0, disturbance secondary	yes
Both maps detect most reserve old-growth, neither all	reserve, Big Reed, Baxter validation (Wilson CIs)	Big Reed 72 [52,86] vs 80 [61,91]; reserves 67 [64,70] vs 57 [54,60]	yes